Phase 31: Final Interview Cheat Sheet (Quick-Recall Reference)

Document Code: SHT-REC-001

Project: Duplicate Bin (`duplicate-bin`)

Purpose: 2-Minute Pre-Interview Memory Refresher

1. Core Project Facts at a Glance

Anchor Dimension	Rapid Recall Fact	Verifiable Code Citation
Project Name	**Duplicate Bin** (`DuplicateBin`)	README.md#L1
Current Versions	Desktop Engine `v0.2.0`; SaaS Backend `v1.0.0`	backend/app/main.py#L18
Business Domain	Storage Optimization, Digital Asset Governance & Hybrid SaaS	docs/SAAS_ANALYSIS_AND_STRATEGY.md
Tech Stack (Desktop)	Python 3.8+, PySide6 (Qt6), SQLite (WAL), Hashlib	requirements.txt
Tech Stack (SaaS)	FastAPI, Uvicorn, SQLAlchemy, WebSockets, JWT Auth	backend/app/main.py
Core Algorithms	Two-pass size pre-filter, SHA-256/BLAKE2b streaming, Union-Find clustering	duplicate_bin/core/detection.py
Safety Innovation	Non-destructive quarantine archive (`~/.duplicate_bin/`) + 1-click restore (`Ctrl+Z`)	duplicate_bin/engine/archive.py
Monetization Model	10 GB Free tier + Pay-As-You-Go Credits ($4.99/100GB) + Team Tiers ($29-$99/mo)	backend/app/models.py#L13

Anchor Dimension	Rapid Recall Fact	Verifiable Code Citation

2. The 30-Second Business Problem & Solution Pitch

- **The Problem:** Unstructured duplicate files consume 15% to 35% of storage across creative prosumers and SME businesses, triggering expensive cloud tier upgrades. Users avoid cleaning because manual review takes hours and they fear accidentally deleting master files.
- **The Solution:** Duplicate Bin delivers an ultra-fast local detection engine (10,000 files in 18.7s) that groups duplicates via Union-Find, applies 8 automated selection heuristics ("Keep Best Quality"), and safely quarantines files with instant `Ctrl+Z` undo. It links to a central cloud dashboard via 6-digit device pairing and WebSockets.

3. Verified Performance & Empirical Benchmarks

- **Initial Cold Scan (10k files): 18.7 seconds** (Standard multi-core SSD).
- **Persistent Cached Rescan (10k files): 4.2 seconds** ($\mathbf{4.45\times}$ faster / **77.54% time reduction**).
- **High-Scale Performance (100k files):** Cold scan 222s down to **28.4s** cached ($\mathbf{7.82\times}$ faster / **87.21% time reduction**).
- **Client RAM Footprint:** Bounded at **180 MB** (10k files) and **850 MB** (100k files) using adaptive 4KB-1MB streaming chunk buffers.
- **Data Loss Rate: 0.0%** (100% verified restoration via JSON quarantine manifests).

4. Key Architectural Patterns to Mention

1. **Two-Pass Size Pre-Filter:** Pass 1 groups files by exact byte size in $O(n)$ time. Unique sizes are discarded immediately; only size collisions proceed to cryptographic hashing.
2. **Union-Find Clustering:** Disjoint-set data structure with path compression clusters transitive duplicates ($A=B, B=C \implies {A,B,C}$) in $O(n \alpha(n))$ near-linear time with zero group fragmentation.
3. **SQLite WAL Hash Caching:** SQLite initialized with `PRAGMA journal_mode=WAL` decouples concurrent

3. **SQLite WAL Hash Caching:** SQLite initialized with `PRAGMA journal_mode=WAL` decouples concurrent worker reads and writes, avoiding database locks.
4. **Zero-Trust Hybrid Relay:** File contents never leave the client machine. The FastAPI backend only handles authentication, 6-digit device pairing, and high-level telemetry metrics (`bytes_recoverable`).
5. **Idempotent Credit Deduction:** Credits are debited atomically in `POST /api/scans/telemetry` strictly upon the initial state transition to `completed`.

5. Core BA Deliverables You Owned

- **BRD & FRD:** Business Requirements Document and Functional Specifications with Gherkin acceptance criteria across all 9 major modules.
- **PRD:** Product vision, target personas ('Elena' & 'Marcus'), MoSCoW backlog prioritization, and release strategy.
- **Process Flows & Gap Analysis:** As-Is manual chaos vs. To-Be automated hybrid workflows modeled in Mermaid diagrams.
- **Data Dictionary & ERD:** Full cloud relational schema (`users`, `devices`, `scans`) and local cache schemas.
- **Traceability & UAT:** Bidirectional RTM tracking 24 formal requirements through unit testing and 10 executed UAT test scenarios with 100% sign-off.

6. Golden Rules for Interview Defense

- **Never invent numbers:** If asked about a metric not in the codebase, say: "*That metric was not part of our empirical baseline and would require telemetry instrumentation to measure.*"
- **Acknowledge the 20x claim:** "*Our marketing cites 20x for high-latency mechanical HDDs, but our empirical SSD benchmark validates a proven 87.21% time reduction (7.82x speedup) on 100,000 files.*"
- **Explain personal contribution:** "*The project implemented the full hybrid system; my role was leading business analysis, requirements modeling, API contract definition, and UAT governance.*"